Qwen3-VL-4B, round 1: broad training, a reasoning delta, and weight averaging

Results of the one-night 4B run (Phase A on all 117 datasets, Phase B on a new hard-reasoning set), the two A/B weight averages, and the same evaluation suite as the 30B and 2B: JevBench public, zero-shot, never-seen tasks, held-out wording, robustness and latency. With a diagnosis of what worked, what did not, and the round-2 plan.

Sources: `results/experiments.jsonl` (every number below is a logged record), JevBench outputs in `results/pod/*/jevbench/`, pod logs in `data/logs/pod/`. Reference values for Imajev-4B and Jev 1.13.0 from the JevBench v1.4.2.2 results file. Code: laya-vision commits 8c3d5f2 (training changes, hard sets), 1dbb140 (generator audit fix), 0d0c31c (soups), ca67d0d (eval pod). Local runs of the soups on the 16 GB GPU were invalid (out of memory at 4,096 tokens) and are excluded.

1. Summary

76.2%	55.0%	82.3%	86.1%
best 4B on JevBench public (4B soup 75% B)	its hard tier (Imajev-4B 72.1%)	our 30B, same items	Imajev-4B, same items

System	All (231)	Easy (48)	Original (72)	Hard (111)	ECE	p50 ms
2B (Qwen3-VL-2B, mix_v5)	60.6	100.0	63.9	41.4	0.102	CPU
30B (Qwen3-VL-30B-A3B, v5)	82.3	100.0	98.6	64.0	0.061	227
4B Phase A	74.5	100.0	95.8	49.5	0.062	91
4B Phase B	76.6	100.0	91.7	56.8	0.055	82
4B soup 50/50	76.2	100.0	95.8	53.2	0.054	89
4B soup 75% B	76.2	100.0	93.1	55.0	0.046	87
Imajev-4B (reference)	86.1	100.0	98.6	72.1	0.026	—
Jev 1.13.0 (reference)	86.6	100.0	~99	—	—	—

Verdict: ship 4B soup 75% B. The four 4B checkpoints are tied outside JevBench (zero-shot, never-seen, held-out wording and MMLU agree within 0.6 points), and on JevBench every one of 4B Phase B, 4B soup 50/50, 4B soup 75% B is within one item of the best; among those, 4B soup 75% B has the lowest calibration error (0.046), balances the original (93.1%) and hard (55.0%) tiers, and has the fewest option-order flips. Across the shared suite (section 4) the 4B matches the 30B on zero-shot accuracy and held-out wording, is better calibrated, and answers an image decision about six times faster; the 30B keeps its lead on knowledge (MMLU) and on the hard tier. The remaining gap to Imajev is almost entirely the hard tier.

2. What was run

Stage	Data	Settings	Hardware / time
Phase A (`qwen4b-v6a`)	mix_v6: all 117 trainable datasets (photos, documents, UI, code incl. execution-verified, support, safety, reasoning, robotics, Jev-style laya_typed); `describe_labels` augmentation 0.3	Qwen3-VL-4B-Instruct, LoRA r32 on attention + MLP, lr 5e-05, batch 16, text budget 4096 tokens (middle cut), logit cap 10.0, CE + RLCD (CE constant)	1× H100, 480 min cap, ~15k steps (~8 rows/s; ≈20% of one pass); best checkpoint step 13,500 by calibration NLL
Phase B (`qwen4b-v6b`)	mix_v6_hard: exact generators (dates / time zones / DST, invoices / discounts / proration, probability), BIG-bench reasoning, ProofWriter, QuALITY, WANLI, HelpSteer2 (~60%); laya_typed; replay of the Phase A domains incl. images	continued from Phase A, lr 2e-05, same budget and loss	60 min cap, 484 steps (long rows)
Soups	—	element-wise average of every saved tensor: 0.5·A + 0.5·B and 0.25·A + 0.75·B	built locally
Evals	JevBench public at 4,096 tokens (their runner and scorer); 14 zero-shot sets; never-seen text / vision; held-out wording (19.7k); robustness (11 sets × 100 × 5 perturbation families); latency (100 image decisions)		1× H100 per checkpoint, ~40 min

Cost: training pod ≈ $45 (12.8 h × $3.49/h, incl. evals of A and B); soup-eval pod ≈ $5 (H100, ~1.4 h incl. upload). Round-1 total ≈ $50.

3. JevBench public in detail

3.1 Hard tier by family (accuracy %)

Hard family (n)	2B (Qwen3-VL-2B, mix_v5)	30B (Qwen3-VL-30B-A3B, v5)	4B Phase A	4B Phase B	4B soup 50/50	4B soup 75% B
temporal_numeric (15)	7	**33**	7	13	7	13
long_policy (19)	**53**	**53**	32	47	37	42
tradeoff (6)	**50**	**50**	**50**	**50**	**50**	**50**
judge_hard (17)	59	**65**	53	59	59	59
probability (10)	**60**	**60**	**60**	**60**	**60**	**60**
multi_hop (18)	17	**78**	50	61	56	56
adversarial (6)	67	**100**	67	67	67	67
ambiguous (7)	0	43	57	**71**	**71**	**71**
routing_hard (5)	80	**100**	**100**	**100**	**100**	**100**
trap (8)	62	**100**	**100**	**100**	**100**	**100**

Families are sorted by Phase B accuracy. n per family is small (6–19 items), so single-family differences below ~15 points are within noise.

3.2 Estimated JevBench composite (their v1.4 scoring code, public tiers only)

System	Intelligence	Calibration	Speed	Cost (price/1k)	Score
30B (Qwen3-VL-30B-A3B, v5)	76.6	87.7	83.1	46.7 ($0.060)	**60.1**
4B Phase A	65.9	87.7	87.7	59.7 ($0.022)	**73.1**
4B Phase B	68.1	88.9	88.3	59.7 ($0.022)	**74.1**
4B soup 50/50	68.2	89.2	87.8	59.7 ($0.022)	**74.1**
4B soup 75% B	67.8	90.7	88.2	59.7 ($0.022)	**74.3**
Imajev-4B (same public-only basis)	81.7	94.8	90.6	59.7	**79.1**

Intelligence uses easy / original / hard only (the judge and sealed tiers are not runnable locally), so these scores are comparable with each other, not with the official board (Imajev-4B official 67.37 includes the sealed set). Cost uses a provider-style price estimate ($0.022 / 1k for 4B-class, Imajev's board figure; $0.06 assumed for the 30B); at our raw serial GPU time the cost axis drops below the board's penalty threshold (PROBLEMS A2).

4. The shared evaluation suite

4.1 Zero-shot (datasets never trained on; accuracy %, raw)

Zero-shot set (n)	30B (Qwen3-VL-30B-A3B, v5)	4B Phase A	4B Phase B	4B soup 50/50	4B soup 75% B
all (12164)	**75.9**	75.8	**75.9**	75.8	75.8
mmlu (1000)	**79.6**	72.0	71.8	71.6	72.0
humaneval_exec (1000)	**88.7**	84.3	84.5	84.6	84.3
cruxeval_output (260)	**68.8**	57.7	57.7	57.7	55.8
mind2web_element (1274)	**78.6**	73.8	74.1	74.3	74.3
mind2web_operation (1942)	31.3	47.5	**48.9**	47.7	48.3
openimages_presence (1500)	**87.5**	86.4	86.0	86.1	86.1
openimages_count (1500)	83.1	**83.7**	83.5	83.6	83.6
openimages_where (316)	**78.2**	77.2	**78.2**	77.2	77.2
openimages_relation (637)	**91.7**	89.6	90.4	90.3	90.1
openimages_framing (107)	**97.2**	93.5	92.5	93.5	93.5
robot_task_zs (450)	**62.7**	59.3	58.7	59.3	58.4
robot_progress_zs (450)	**77.8**	66.4	64.4	65.8	65.6
robospatial_home (228)	65.8	65.8	**67.1**	66.2	**67.1**
probe_state_facts (1500)	**98.9**	97.4	97.2	97.4	97.3
ECE after temperature (all)	0.066	**0.041**	0.044	0.045	0.043

4.2 Never-seen tasks, held-out wording, latency

Eval	30B (Qwen3-VL-30B-A3B, v5)	4B Phase A	4B Phase B	4B soup 50/50	4B soup 75% B
Never-seen text (1,200)	**71.3**	67.8	67.2	67.0	67.0
Never-seen vision (750)	**92.8**	89.2	89.5	89.2	89.5
Held-out wording, all (19,708)	**91.3**	90.6	90.5	90.5	90.5
· laya_typed (2,000)	80.9	**82.2**	81.6	81.8	81.8
· laya_typed_img (2,000)	76.9	**78.2**	77.4	78.1	77.8
· code_choice (141)	**62.4**	48.9	48.2	48.2	49.6
Latency, image decision, median ms	561	**93**	**93**	95	96
· GPU	H200	H100	H100	H100	H100

4.3 Robustness

Flip rate % (11 sets × 100)	30B (Qwen3-VL-30B-A3B, v5)	4B Phase A	4B Phase B	4B soup 50/50	4B soup 75% B
option_order: option order shuffled (lower is better)	3.1	3.1	3.3	3.5	**3.0**
text: question paraphrased (lower is better)	2.2	1.8	1.7	**1.5**	1.6
image: image perturbed (lower is better)	**2.5**	3.6	3.4	3.5	3.3
image_shuffle: image replaced by another (higher = uses the image)	**58.3**	57.6	56.9	57.1	56.9
text_only: image removed (higher = uses the image)	56.0	56.9	56.7	56.7	**57.1**

For the first three families a lower flip rate is better (the answer should not change); for the last two a higher rate means the model's answer depends on the image, as it should for image questions.

5. Training dynamics

5.1 Phase A

Phase A eval step	Accuracy (30/set, 2,960)	ECE
0 (start)	32.0	0.037
1500	78.6	0.041
3000	80.9	0.013
4500	81.2	0.028
6000	82.8	0.053
7500	83.3	0.027
9000	83.9	0.037
10500	84.4	0.045
12000	84.6	0.036
13500	84.8	0.033
14997	85.0	0.047

5.2 Phase B: what the reasoning delta changed

Phase B eval set (30 rows each)	Start (= Phase A)	End (step 484)	Δ points
gen_temporal	40.0	40.0	+0.0
gen_money	13.3	43.3	+30.0
gen_probability	66.7	80.0	+13.3
bb_reasoning	73.3	83.3	+10.0
proofwriter_owa	50.0	66.7	+16.7
wanli_nli	66.7	66.7	+0.0
quality_long	76.7	80.0	+3.3
helpsteer_score	46.7	46.7	+0.0
helpsteer_pair	76.7	86.7	+10.0
laya_typed	56.7	60.0	+3.3
laya_typed_img	66.7	66.7	+0.0
replay sets, mean of 63 (images, docs, code, support, robots)	84.8	84.9	+0.1

30 validation rows per set (±9 points each); read the direction, not the exact size.

6. Findings

1. **The 4B is the right product tier.** On everything except knowledge and the hard tier it matches the 30B, with better calibration and ~6× lower latency; on the estimated composite it beats the 30B, whose cost axis is penalised.
2. **The format fix worked.** The original tier went from 63.9% (2B) to 95.8% (Phase A); `describe_labels` plus the 4,096-token middle-cut budget is the likely cause (not separately ablated).
3. **Phase B buys reasoning and costs format.** Hard +7.3 points (multi_hop, ambiguous, long_policy up), original −4.1 (3 of 72 items): the forgetting Imajev reported. **Weight averaging trades the two smoothly**: 50/50 restores the original tier (95.8%) and keeps half the hard gain (53.2%); 75% B sits between (93.1% / 55.0%) with the best calibration of any 4B (ECE 0.046). All four checkpoints are equal on JevBench within one item.
4. **Dates and time zones did not move** (temporal_numeric 7–13%): ~500 Phase B steps over 3 generator kinds are not enough for multi-step calendar arithmetic in one pass.
5. **Option-order flips are unchanged** (~3%): an architecture problem (per-option read-out on a causal backbone), not data.
6. **Coverage was thin:** Phase A saw ~20% of one pass; long-tail datasets got ~2k draws each.

7. Round 2 (ordered by expected gain per dollar)

#	Action	Targets	Cost
1	Ship the best soup; refit temperatures per type × option count on held-out Jev-shaped items	ECE → ~0.03; calibration axis +5	local
2	Generators: the 4 missing families (unit conversion, rule precedence, multi-hop tables, withheld input → unknown) + multi-step compositions (business days + DST + grace in one question), 3–5× more temporal rows; long generated policies (1.5–3.5k tokens)	temporal_numeric, long_policy, multi_hop	CPU
3	Teacher-lite: ~3k documents → ~9k questions, 2 open-weight thinking answerers, keep on agreement, soft targets, rationales	judge_hard, tradeoff, ambiguous	≈ $25
4	Phase B2: 2–3 h from the shipped soup, rationale loss, **35–40% replay** (Phase B's ~22% cost 4 original-tier points; CLM kept 69.0 → 68.5 with 40%), agentic trajectory domain (V6.2 §4); averaging sweep; gates	hard 65–70% (estimate)	≈ $35–50
4b	Agentic trajectory builder (V6.2 §4): next action, step progress, outcome on the value head, verifier pairs; licence review per source; best-of-N eval against random / oracle	calibrated agent-run judge	CPU, 2–3 days
5	Abstain + reason heads on the existing act-head features (V6.1 §6.2)	calibrated "cannot decide"	≈ 1 day
6	Block option attention + shared option positions on Qwen	flips < 1.5%	≈ 2 days
7	Serving: merged LoRA, fp8, CUDA graphs, Qwen prefix cache; hosted endpoint for the sealed JevBench / Image JevBench	cost axis, official rank	engineering

Appendix: record ids

Record	id	time
hof:qwen30b-v5	20260928-185811-f2d54a	2026-09-28T18:58
hof:qwen4b-v6a	20260929-153930-0fc725	2026-09-29T15:39
hof:qwen4b-v6b	20260929-161754-8b7282	2026-09-29T16:17
hof:qwen4b-v6b-soup50	20260929-173002-82aaec	2026-09-29T17:30
hof:qwen4b-v6b-soup75	20260929-180352-f2cdba	2026-09-29T18:03
jevbench-public:qwen2b-mix	20260928-143154-5c8559	2026-09-28T14:31
jevbench-public:qwen2b-mix-len4096	20260928-221417-e94a75	2026-09-28T22:14
jevbench-public:qwen30b-v5	20260929-014950-0fd688	2026-09-29T01:49
jevbench-public:qwen30b-v5-len4096	20260929-020813-a50bbe	2026-09-29T02:08
jevbench-public:qwen4b-v6a-len4096	20260929-151641-144712	2026-09-29T15:16
jevbench-public:qwen4b-v6b-len4096	20260929-155619-80265f	2026-09-29T15:56
jevbench-public:qwen4b-v6b-soup50-len4096	20260929-170849-0f7568	2026-09-29T17:08
jevbench-public:qwen4b-v6b-soup75-len4096	20260929-174300-33dc09	2026-09-29T17:43
latency-qwen30b-v5	20260928-193015-8b30df	2026-09-28T19:30
latency-qwen4b-v6a	20260929-155445-4348db	2026-09-29T15:54
latency-qwen4b-v6b	20260929-163244-bf956a	2026-09-29T16:32
latency-qwen4b-v6b-soup50	20260929-174225-6109af	2026-09-29T17:42
latency-qwen4b-v6b-soup75	20260929-181553-a86d81	2026-09-29T18:15
laya-full:qwen30b-v5	20260929-015536-4bf13a	2026-09-29T01:55
mix-split1024->qwen2b-mix	20260925-152638-951397	2026-09-25T15:26
novel-v1:qwen2b-mix	20260928-122122-f6810d	2026-09-28T12:21
novel:qwen30b-v5	20260929-015859-2006be	2026-09-29T01:58
novel:qwen4b-v6a	20260929-152704-c567ca	2026-09-29T15:27
novel:qwen4b-v6b	20260929-160610-652f2b	2026-09-29T16:06
novel:qwen4b-v6b-soup50	20260929-171808-99c6fc	2026-09-29T17:18
novel:qwen4b-v6b-soup75	20260929-175156-c0c992	2026-09-29T17:51
overfit-gap:qwen30b-v5	20260928-221347-964f91	2026-09-28T22:13
overfit-train:qwen30b-v5	20260929-020440-970f36	2026-09-29T02:04
quant-v1:qwen2b-mix	20260927-041230-24e217	2026-09-27T04:12
qwen2b-mix	20260925-150248-d45322	2026-09-25T15:02
qwen30b-v5	20260928-180343-0d7fd8	2026-09-28T18:03

qwen30b-v5-smoke	20260927-233751-e8d82d	2026-09-27T23:37
qwen4b-v6a	20260929-132637-271e73	2026-09-29T13:26
qwen4b-v6a-smoke	20260929-043316-73c41b	2026-09-29T04:33
qwen4b-v6b	20260929-151326-5d7901	2026-09-29T15:13
qwen4b-v6b-soup50	20260929-123956-46039f	2026-09-29T12:39
qwen4b-v6b-soup75	20260929-123957-c867c3	2026-09-29T12:39
robust-v1:qwen2b-mix	20260927-051334-bfb2ed	2026-09-27T05:13
robust:qwen30b-v5	20260928-192700-f4593e	2026-09-28T19:27
robust:qwen4b-v6a	20260929-155232-6a6949	2026-09-29T15:52
robust:qwen4b-v6b	20260929-163050-186a0b	2026-09-29T16:30
robust:qwen4b-v6b-soup50	20260929-174143-83e5b1	2026-09-29T17:41
robust:qwen4b-v6b-soup75	20260929-181512-002abd	2026-09-29T18:15
zeroshot-oi-m2w:qwen2b-mix	20260925-151211-07ae99	2026-09-25T15:12
zeroshot-v2:qwen2b-mix	20260927-033149-2da048	2026-09-27T03:31
zeroshot:qwen30b-v5	20260928-182532-1df751	2026-09-28T18:25
zeroshot:qwen4b-v6a	20260929-152515-1d33b7	2026-09-29T15:25
zeroshot:qwen4b-v6b	20260929-160441-82b9b6	2026-09-29T16:04
zeroshot:qwen4b-v6b-soup50	20260929-171717-151f4a	2026-09-29T17:17
zeroshot:qwen4b-v6b-soup75	20260929-175106-db6132	2026-09-29T17:51